Russell W. Mercer

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Education

Brigham Young University, Provo, UT — January 2024 – April 2028 (Expected)

B.S. in Biochemistry, Minor in Computer Science

- Cumulative GPA: 3.99; Science GPA: 3.99
- Advanced biochemistry, pre-medical and machine learning coursework

Medical College Admission Test (MCAT): 519, 96th percentile — August 2026

Research Experience

Biochemistry Research Assistant, Jensen Lab, BYU — August 2026 – Present

Department of Chemistry and Biochemistry; Mentor: Grant J. Jensen, PhD (Dean, College of Physical and Mathematical Sciences); with Chitral Chatterjee

- Testing whether a general-purpose AI coding agent (Claude Code) can shorten the cryo-electron tomography learning curve for a new researcher without replacing the understanding the work normally builds
- Processing 39 tilt series of cryo-FIB-milled, exponentially growing *E. coli* through motion correction, alignment, reconstruction, missing-wedge correction, template matching, classification and subtomogram averaging, ending in a ribosome neighbor analysis of disomes and polysomes
- Recording which agent recommendations are accepted, overridden or caught as errors, and packaging the workflow as a reusable agent skill
- Completed Thermo Fisher Scientific "Resolution Evolution" cryo-EM training with hands-on Vitrobot and Krios use

Clinical Research Assistant, UT Southwestern Medical Center — May 2026 – Present

Radiation Oncology, Kenneth Westover, MD, PhD (May – August 2026); Medical Physics, Jie Deng, PhD (August 2026 – Present)

- Started through the BYU Summer Premedical Research Internship, funded by the BYU Simmons Center for Cancer Research; worked on site in Dallas, TX through August 2026 and remotely since
- Contributing to studies of lung and pancreas diffusion-weighted MRI on the Elekta Unity MR-Linac, biology-guided radiotherapy (RefleXion) and radiolucent patient immobilization (Aerframe)
- Abstracted clinical data in Epic, prepared figures and drafted manuscripts
- Offered a continuing position in Medical Physics by Dr. Deng after the summer internship

Chemistry Research Assistant, Stowers Lab, BYU — April 2024 – May 2026

Department of Chemistry and Biochemistry; Mentor: Kara J. Stowers, PhD

- Synthesized surface-bound metal-organic framework thin films (CuBDC/DABCO, HKUST-1) by spin coating and studied their structure and decomposition
- Helped build, repair and maintain a flow reactor and an ultra-high-vacuum chamber with XPS and mass spectrometry, including leak checks and bakeouts, reaching about 10^{-10} Torr
- Imaged films and measured film thickness by SEM (Thermo Fisher Verios G4, Helios) and ImageJ
- Trained newer lab members in spin coating and SEM imaging
- Supported by a BYU Undergraduate Research Award; still contributing to a manuscript in preparation

Publications

Peer-Reviewed

Ralph AD, **Mercer RW**, Hall SB, Butikofer BC, Collins CS, Sturgill CJ, Minson PS, Stefik M, Stowers KJ. Ambient pressure fabrication of $Cu_xZn_{1-x}(BDC)_2DABCO$ thin films through spin-coating and their temperature-dependent structural development. *Langmuir*. 2026;42(6):4475–4488. doi:10.1021/acs.langmuir.5c05182

In Preparation

[TODO: authors and title] **Mercer RW**, et al. Lung diffusion-weighted MRI on the Elekta Unity MR-Linac (UT Southwestern). *Manuscript in preparation.*

[TODO: authors and title] **Mercer RW**, et al. Biology-guided radiotherapy with RefleXion (UT Southwestern). *Manuscript under co-author review.*

[TODO: authors and title] **Mercer RW**, et al. Radiolucent patient immobilization with Aerframe (UT Southwestern). *Manuscript under co-author review.*

[TODO: authors and title] **Mercer RW**, et al. Pancreas diffusion-weighted MRI on the Elekta Unity MR-Linac (UT Southwestern). *In progress.*

[TODO: authors and title] **Mercer RW**, et al. Using an AI coding agent to learn cryo-electron tomography processing (Jensen Lab, BYU). *In preparation.*

[TODO: authors, title and topic] **Mercer RW**, et al. Co-author, UT Southwestern study. *In preparation.*

[TODO: authors, title and topic] **Mercer RW**, et al. Co-author, UT Southwestern study. *In preparation.*

[TODO: authors and title] **Mercer RW**, et al. Co-author, surface chemistry study using the Stowers Lab UHV chamber (BYU). *In preparation.*

Presentations and Abstracts

Mercer RW, Chatterjee C, Jensen GJ. Can an AI agent collapse the cryo-ET learning curve? Poster, Utah CryoEM Symposium, Brigham Young University, Provo, UT. October 2026

[TODO: title]. Poster, BYU Medical Society, Brigham Young University, Provo, UT. August 2026

[TODO: title]. Oral presentation, BYU Simmons Center for Cancer Research, Provo, UT. August 2026

[TODO: title]. Poster, American Chemical Society Spring Meeting, Atlanta, GA. March 2026

[TODO: title]. Oral presentation, BYU Student Research Conference, Provo, UT. February 2026

Fast room temperature synthesis and characterization of a surface bound metal organic framework without the use of a self-assembling monolayer. Presentation, BYU Student Research Conference, Provo, UT. March 2025

Honors, Awards and Funding

SCCR Fellow, BYU Simmons Center for Cancer Research (funded summer research at UT Southwestern) 2026

Outstanding Analytical Chemist Award, BYU 2025

Dean's List, BYU (Winter, Spring and Fall 2024; Fall 2025) 2024 – 2025

Undergraduate Research Award, BYU (awarded each semester) 2024 – 2026

Stoney Black Flight of the Phoenix Award, Air Academy High School Marching Band 2021

Eagle Scout, Boy Scouts of America 2020

Clinical Experience

Medical Assistant, Health Plus Clinic, Springville, UT — July 2025 – April 2026

- Assisted a D.O. and a P.A. with patient care and clinical procedures
- Kept medical records accurate and compliant; completed HIPAA and medical assistant training

Volunteer, Utah State Hospital, Provo, UT — May 2025 – Present

- Physical Therapy (May 2025 – Present): guide 10+ mental health patients through therapeutic exercise and help the physical therapist keep sessions safe and encouraging
- Pediatric Center (September 2025 – January 2026): led group activities for pediatric mental health patients
- Pediatric Mentor (December 2025 – May 2026): met one-on-one with pediatric patients

Physician Shadowing

Dan Baer, Pain Management (PM&R), 3.5 hours — July 2025
Michael Benson, Psychiatry, Utah State Hospital, 8 hours — August 2025
Matheson Harris, Plastic Surgery (Facial Reconstruction), Utah Surgical, 7.5 hours — November 2025
Stanford C. Taylor, Retina Surgery, Pueblo, CO, 6 hours — April 2026
Thomas Fowler, Allergy, 4 hours — June 2026
Joshua Jaramillo, Trauma Surgery and ICU, Utah Valley Hospital **[UPCOMING: add hours after]** — Nov – Dec 2026

Leadership and Teaching

Officer, BYU AI Association — September 2026 – Present

- Lead the weekly AI in Research meeting and wrote the proposal for it

Pre-Med Peer Mentor, BYU Pre-Med Association — October 2026 – Present

- Mentor pre-med students on coursework, research and clinical experience

Lead Program Director, BYU Y-Serve, Early Learning Essentials — August 2024 – Present

- Program Director from August 2024; Lead Program Director since March 2025
- Recruit volunteers, organize events and manage communication for a program serving preschoolers from low-income families

Peer Mentor, BYU Office of First Year Experience — June 2024 – February 2025

- Co-taught two sections of Introduction to University (UNIV 101) with faculty
- Mentored 45 first-year students through regular one-on-one meetings; completed QPR gatekeeper training

Full-Time Missionary, California Riverside Mission — October 2021 – December 2023

The Church of Jesus Christ of Latter-day Saints

- Led and supported small groups of fellow volunteers
- Planned and gave 80+ trainings on teaching skills to groups of 8 to 100 volunteers; mentored four new volunteers one-on-one
- Worked in the mission office coordinating housing and travel logistics and managing technology

Squad Leader, Air Academy High School Marching Band, Colorado Springs, CO — May 2020 – May 2021

- Led the trumpet section; the band won two state championships and marched in the 2018 Rose Parade

Service

Volunteer, Global Early Detection Initiative, Utah Chapter (community outreach) — August 2026 – Present
Church Instructor, The Church of Jesus Christ of Latter-day Saints — August 2026 – Present

Congregation Activities Committee and Leadership Council — September 2025 – April 2026
Congregation Clerk (records) and Leadership Council — May 2024 – August 2024
Boy Scouts of America; Eagle Scout project for the National Mill Dog Rescue — 2014 – 2020

Certifications and Training

NHCO Certified Phlebotomy Technician — June 2024
Thermo Fisher Scientific "Resolution Evolution" Cryo-EM Training (Vitrobot, Krios) — October 2026
Epic for Research Coordinators, UT Southwestern — May 2026
SCCR Research Ethics Seminar, BYU — April 2026
QPR Suicide Prevention Gatekeeper Training, BYU — 2024

Conferences Attended

AI for Social Innovation Conference **[UPCOMING: Oct 21]** — October 2026
Utah CryoEM Symposium, Brigham Young University — October 2026
American Chemical Society Spring Meeting, Atlanta, GA — March 2026

Skills

Laboratory: SEM, ultra-high vacuum systems, XPS, mass spectrometry, spin coating, flow reactor operation, Vitrobot, Krios cryo-TEM, phlebotomy

Computational: Python, Java, ImageJ, Epic, cryo-ET processing software; AI tools since 2024, with strong working knowledge of Claude Code and building agent workflows that work with data and computer systems

Activities

BYU Fencing Club — August 2026 – Present
Intramural Ultimate Frisbee, BYU — May 2025 – December 2025